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Reliable Smartphone-Based Detection of Counterfeit Bangladeshi Banknotes: Multi-View Deep Learning with Out-of-Distribution Rejection and a Benchmark Audit

Abstract: Objective: Counterfeit high-value banknotes harm retailers and individuals who cannot afford verification hardware. This study develops a reliable smartphone-based authenticator for Bangladeshi 500-taka and 1000-taka notes and examines whether the public benchmark used to evaluate such systems can still distinguish between methods. **Methodology:** The system fine-tunes an EfficientNet-B0 network, fuses up to six photographs of each note, returns a genuine, uncertain, or counterfeit verdict, rejects unsupported inputs with an out-of-distribution gate, and explains each decision with gradient-weighted class activation maps. It was evaluated on the JaalTaka dataset (1,390 notes: 802 genuine and 588 seized counterfeits, six photographs each), which was also audited for duplicate notes. **Findings:** On 208 held-out test notes, the system reached 98.08% accuracy (95% confidence interval 95.16–99.25%) and an area under the receiver operating characteristic curve of 0.994, taking 37 milliseconds per photograph on an ordinary computer processor. However, the 1,390 notes form only 846 distinct photograph sets, 116 test notes have a copy in the training data, and 12 duplicate groups carry both labels, causing all four test errors. On the 92 test notes without a copy, the system and two simple baselines were error-free. **Conclusions:** The benchmark is saturated and can no longer rank methods. The duplicate groups are made available, and group-aware evaluation that holds out counterfeit families is recommended.

Keywords: *Taka authentication; EfficientNet; Grad-CAM; data leakage; explainable AI; transfer learning.*

1. Introduction

Counterfeit Tk 500 and Tk 1000 notes undermine trust in cash, the main means of payment for small retailers in Bangladesh. Banks use ultraviolet, infrared, and magnetic sorters, but shopkeepers and citizens rely on visual and tactile checks that are slow and unreliable for untrained users. A smartphone camera, carried by almost everyone, makes image-based authentication an attractive complement.

The JaalTaka dataset [1] enabled such work for Bangladeshi notes by releasing smartphone photographs of genuine notes and of counterfeits seized by the Rapid Action Battalion (RAB); its authors reported 97.4–98.9% accuracy for five ImageNet convolutional neural networks (CNNs). High benchmark accuracy is necessary but not sufficient for a tool that people trust with money. It must also (i) handle unsupported inputs, (ii) admit uncertainty, (iii) explain its decisions, and (iv) be evaluated in a way that predicts behaviour on unseen notes. The last point is easy to get wrong: networks exploit shortcuts [2], and duplicates across splits inflate accuracy [3]. Our contributions are:

- an open pipeline (Fig. 1) that runs in real time on a central processing unit (CPU) and combines EfficientNet-B0 [4], fusion of up to six photographs, a three-way decision with adjustable strictness, an out-of-distribution (OOD) gate, and gradient-weighted class activation mapping (Grad-CAM) [5];
- a note-disjoint evaluation with confidence intervals, threshold and OOD analyses, and frozen-feature baselines; and
- a duplicate audit showing that label conflicts between copied notes cause every test error and that the benchmark is saturated, plus an occlusion test revealing signal outside the note's centre.

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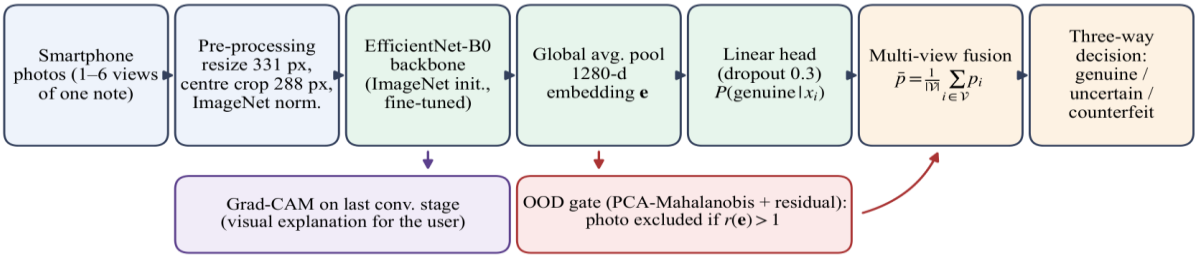


Fig. 1. Overview of the system. Each photograph is classified by a fine-tuned EfficientNet-B0; photographs failing the OOD test are excluded, the remaining genuine probabilities are averaged, and a three-way verdict is returned with Grad-CAM maps.

2. Related Work

Early detectors used hand-crafted features: luminance histograms with multiple-kernel support vector machines (SVMs) [6], scanned background printing texture [7], the multispectral response of inks [8], and individual security features of Bangladeshi notes [9]; a mobile application was also proposed [10]. These methods depend on controlled imaging and denomination-specific tuning. Transfer learning from ImageNet [11,12] lets CNNs learn from few images: Pham et al. detected fake banknotes in smartphone images [13,14], Bangladeshi denominations were recognised with neural networks and lightweight CNNs [15,16], JaalTaka was benchmarked [1] with DenseNet121 [17], ResNet50 [18], VGG16 [19], MobileNetV2 [20], and InceptionV3 [21], and transfer learning was also applied in [22]. None of these studies evaluates OOD inputs, uncertainty, or duplicates.

Maximum softmax probability is a weak OOD signal [23]; Mahalanobis distances on network features are stronger [24], and we adapt them to a low-dimensional subspace. Miscalibrated probabilities [25] motivate an explicit "uncertain" outcome, and Grad-CAM [5] shows the regions supporting a class. Shortcut learning [2] and train–test leakage [3] motivate our audit.

3. Materials and Methods

3.1. Dataset

JaalTaka [1,26] (CC BY 4.0) contains 1,390 notes: 802 genuine (398 Tk 500, 404 Tk 1000) from three national banks and 588 counterfeit (320 Tk 500, 268 Tk 1000) supplied by RAB. Each note was photographed six times, showing different segments (Fig. 2), with one of four smartphones (Samsung Galaxy A50 and M31, Redmi Note 8 and Note 9 Pro), giving 8,340 images, which we resized to a longer side of 1,024 pixels. According to the dataset description, 649 original notes (502 genuine, 147 counterfeit) were expanded to 1,390 by augmentation.

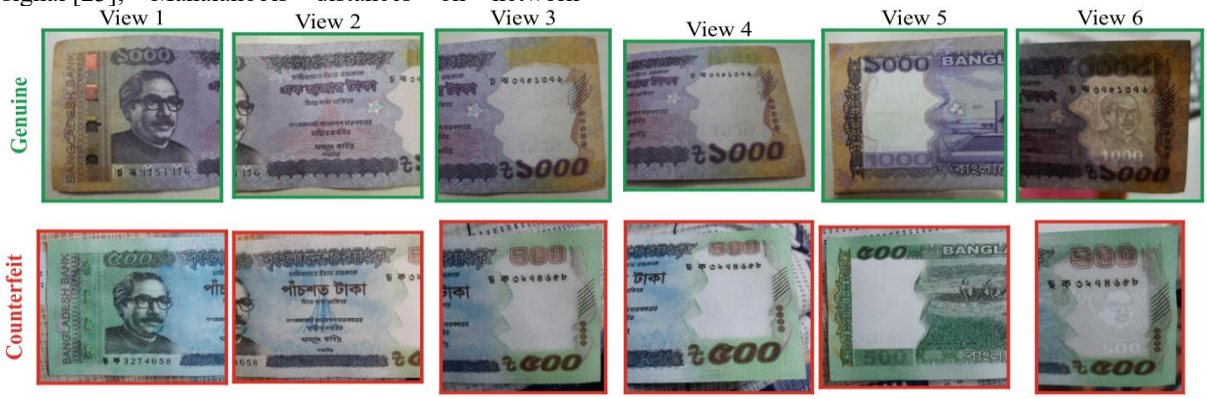


Fig. 2. The six photographs of a genuine Tk 1000 note (top) and a counterfeit Tk 500 note (bottom) from the test split; colour, lighting, and background vary with the phone and session.

3.2. Evaluation Protocol

Because the photographs of a note are correlated, we split by note: within each class, notes were shuffled (seed 42) and assigned 70/15/15% to training (562 genuine, 412 counterfeit; 5,844 photographs), validation, and test (120 genuine, 88 counterfeit; 1,248 photographs each). Validation serves model selection and OOD calibration; the test split is used once. Section 4.6 shows that this note-disjoint split is *not* sufficient, because distinct notes can share photographs. Genuine is the positive class. We report accuracy with a 95% Wilson interval [27], the counterfeit detection rate (specificity), the genuine acceptance rate (recall), and the area under the receiver operating characteristic curve (ROC-

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AUC) [28], per photograph (n = 1,248) and per note (n = 208), at threshold 0.5 unless stated otherwise.

3.3. Classifier and Training

EfficientNet-B0 [4], chosen for its accuracy–cost trade-off on CPUs and initialised with ImageNet weights [11], maps a 288×288 input x to a 1280×9×9 feature map; global average pooling yields the embedding $\mathbf{e} = f(x) \in \mathbb{R}^{1280}$. The head is dropout ($p$ = 0.3) and a linear layer:

$$p(x) = \mathrm{softmax}(We + b)_{\text{genuine}} \tag{1}$$

The model has 4.01 M parameters. Training uses random resized crops (scale 0.35–1.0, aspect 0.75–1.33), rotations up to ±10°, colour jitter (brightness and contrast ±0.4, saturation ±0.3, hue ±0.04), and Gaussian blur (p = 0.2). At evaluation, the shorter side is resized to 331 pixels and the central 288×288 crop is kept. The stem and first three MBConv stages are frozen with their ImageNet batch-norm statistics; the rest is fine-tuned with the class-weighted, label-smoothed [21] cross-entropy

$$\begin{aligned} \mathcal{L} &= -\textstyle\sum_{c\in\{0,1\}} w_c q_c \log \hat{y}_c\,, \\ q_c &= (1-\varepsilon)\mathbb{1}[c=y] + \tfrac{\varepsilon}{2} \end{aligned} \tag{2}$$

with ε = 0.05 and $w_c = N/(2N_c)$. We use AdamW [29] (weight decay 10^{-4}), a one-cycle schedule [30] (peak 3×10^{-4}, 15% warm-up), batch size 12, and six epochs, keeping the epoch with the highest validation accuracy plus AUC; accuracy rose from 98.80% after the first epoch to 99.04% (AUC 0.9976) after the sixth, which was selected. Training took about 55 min on an Intel Core i5-12400 CPU with PyTorch [31].

3.4. Multi-View Fusion and Three-Way Decision

Let V be the photographs of a note that pass the OOD gate. The note-level probability is the mean, a late-fusion form of multi-view pooling [32],

$$\bar{p} = \tfrac{1}{|\mathcal{V}|}\textstyle\sum_{i\in\mathcal{V}} p(x_i) \tag{3}$$

and, for a strictness threshold $\tau \in [0.5,1)$ (default 0.7), the verdict is

$$\text{verdict} = \begin{cases} \text{not supported} & \text{if } \mathcal{V} = \emptyset \\ \text{genuine} & \text{if } \bar{p} \geq \tau \\ \text{counterfeit} & \text{if } \bar{p} < 0.5 \\ \text{uncertain} & \text{otherwise} \end{cases} \tag{4}$$

Averaging needs no extra training and accepts any number of views. The asymmetry reflects costs: accepting a counterfeit causes a loss, whereas "uncertain" only asks the user to check the note by hand or add photographs. Users can choose $\tau \in \{0.6,0.7,0.85\}$ or any other value.

3.5. Out-of-Distribution Gate

Following [24], let $\boldsymbol{\mu}$ be the mean training embedding and $\mathbf{U}_k$ its first k = 64 principal directions with variances σ_j^2. With $\mathbf{z} = \mathbf{U}_k^{\top}(\mathbf{e}-\boldsymbol{\mu})$, we compute the in-subspace Mahalanobis distance and the out-of-subspace residual

$$\begin{aligned} m(e) &= \textstyle\sum_{j=1}^{k} \frac{z_j^2}{\sigma_j^2}, \\ \rho(e) &= \|(e-\mu) - U_k z\|_2^2 \end{aligned} \tag{5}$$

Thresholds θ_m and θ_ρ are the 99.5th validation percentiles. A photograph is accepted when $r(\mathbf{e}) = \max\{m/\theta_m,\ \rho/\theta_\rho\} \leq 1$ and its Laplacian variance (a texture measure) is at least 30, which rejects blank and blurred images. The residual catches embeddings that leave the banknote subspace, which the in-subspace distance alone cannot detect.

3.6. Explanation and Deployment

Grad-CAM [5] is computed on the last feature map $A \in \mathbb{R}^{1280\times9\times9}$ for the predicted class c:

$$\begin{aligned} \alpha_k &= \tfrac{1}{81}\textstyle\sum_{u,v} \frac{\partial y^c}{\partial A_{uv}^k}, \\ L^c &= \mathrm{ReLU}(\textstyle\sum_k \alpha_k A^k) \end{aligned} \tag{6}$$

where y^c is the logit of class c; the map is normalised, upsampled, and overlaid on the photograph. The model runs in 37 ms per photograph (median of 30 runs, six CPU threads). The deployed Flask and Streamlit interfaces also average a full-frame crop; all reported results use the central crop only.

4. Results

4.1. Main Results and Baselines

Per note, the system rejects 86 of 88 counterfeits and accepts 118 of 120 genuine notes (Table 1, Fig. 3), and both ROC curves in Fig. 4 rise almost vertically. The per-note confusion matrix equals the per-photograph matrix divided by six, and each view position alone reaches 98.08%: every error comes from all six photographs of one of four notes.

Table 1. Test results of the proposed system (threshold 0.5)

Metric	Per photograph	Per note (6 views)
Samples (genuine / counterfeit)	1,248 (720 / 528)	208 (120 / 88)
Accuracy (%)	98.08	98.08
95% Wilson CI (%)	97.15–98.70	95.16–99.25
Counterfeit detection (%)	97.73	97.73 (86/88)
Genuine acceptance (%)	98.33	98.33 (118/120)
Balanced accuracy (%)	98.03	98.03
F1, genuine class	0.983	0.983
ROC-AUC	0.990	0.994

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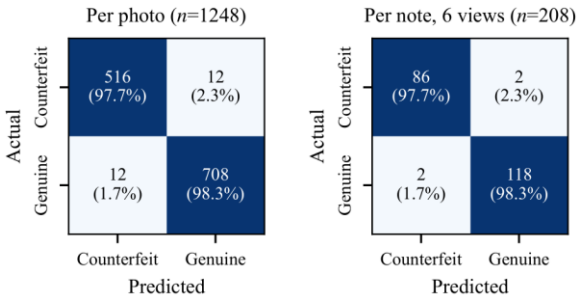


Fig. 3. Test confusion matrices per photograph (left) and per note (right).

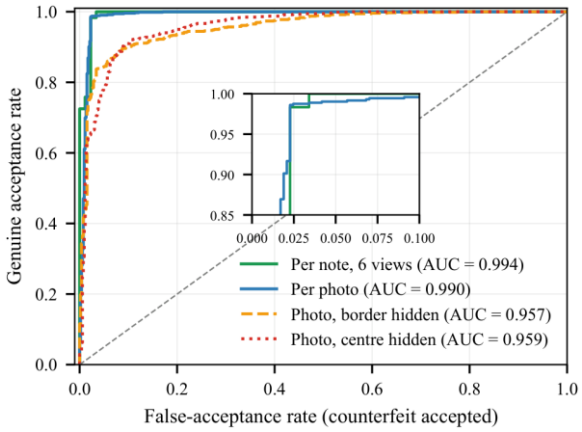


Fig. 4. Test ROC curves for per-note fusion, single photographs, and the occlusion diagnostics of Section 4.5.

We also compared a cosine 1-nearest-neighbour (1-NN) classifier and a logistic-regression probe on *frozen* ImageNet EfficientNet-B0 embeddings (Table 2). Fine-tuning adds 1.4–1.6 points of per-photograph accuracy, but at note level the probe matches it and 1-NN errs on only six notes. The JaalTaka authors' results (97.4–98.9% [1]) used a different split and are not directly comparable.

Table 2. Comparison with frozen-feature baselines on the test split

	Per photograph		Per note	
Method	Acc. (%)	AUC	Acc. (%)	AUC
1-NN, frozen ImageNet features	96.71	0.967	97.12	0.979
Linear probe, frozen features	96.47	0.981	98.08	0.983
Fine-tuned EfficientNet-B0 (ours)	**98.08**	**0.990**	**98.08**	**0.994**

4.2. Fusion and the Strictness Threshold

Over all C(6, k) subsets of k views, accuracy stays at 98.08% while the mean AUC rises monotonically from 0.9900 (k = 1) through 0.9912, 0.9917, 0.9925, and 0.9930 to 0.9939 (k = 6): fusion makes the ranking of notes more reliable. The fused scores are strongly bimodal (Fig. 5): correctly classified genuine notes have $\bar{p} > 0.86$ and counterfeits $\bar{p} < 0.10$, so outcomes are unchanged for $\tau \leq 0.85$, and at $\tau = 0.90$ one genuine note (0.8%) becomes "uncertain". The two accepted counterfeits ($\bar{p} \approx 0.95$) would need $\tau > 0.954$, which would send most genuine notes to manual inspection.

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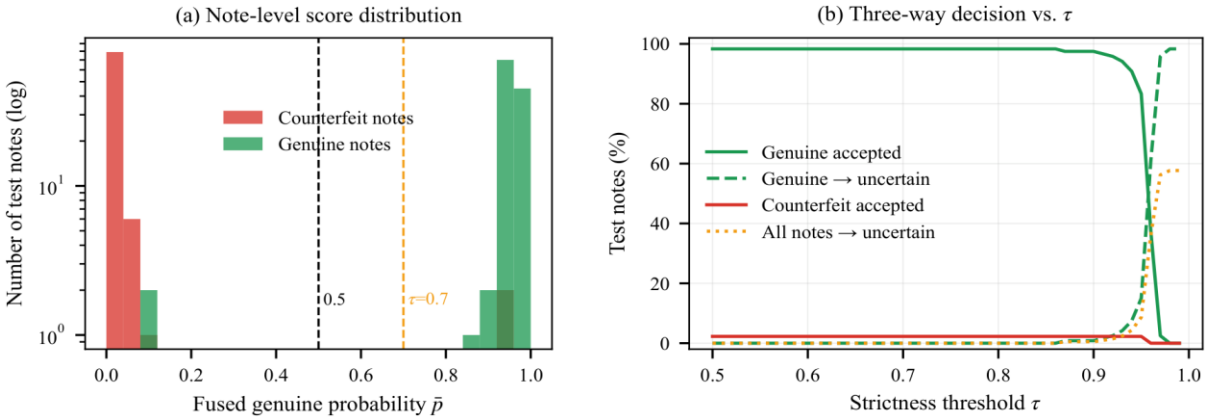


Fig. 5. (a) Fused genuine probability $\bar{p}$ of the 208 test notes. (b) Share of test notes accepted, "uncertain", or wrongly accepted versus the threshold τ.

4.3. Out-of-Distribution Gate

The gate rejects 4 of 1,248 test photographs (0.32%) versus 0.72% on validation, with a median test ratio of 0.26 (Fig. 6); no test note was "not supported". In a preliminary check, it rejected 100 of 105 photographs of other denominations (Tk 2–200) and blank, noise, and unrelated images; a systematic OOD benchmark remains future work.

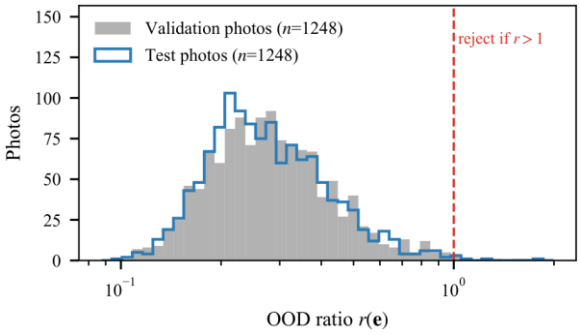


Fig. 6. OOD ratio $r(\mathbf{e})$ of validation and test photographs; the dashed line is the rejection limit.

4.4. Visual Explanations

In Fig. 7, Grad-CAM concentrates on the portrait and the denomination and ornament panels for correct decisions, and also on the security band for the counterfeit in row (b). The wrongly accepted note in row (c) looks genuine to the model, as expected because an identical copy appears in training labelled genuine (Section 4.6).

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Fig. 7. Grad-CAM on test notes (views 1 and 2). (a) Genuine, accepted. (b) Counterfeit, rejected. (c) Counterfeit, wrongly accepted; its photographs occur in training under the genuine label.

4.5. Shortcut Diagnostic

As shown in Fig. 8, we blacked out either the centre (keeping a 20% border) or the border of each test input. Hiding the border gives 88.94% accuracy (AUC 0.957). Hiding the centre gives 75.48%, as most genuine photographs are rejected (acceptance 58.47%), yet the AUC remains 0.959 and 98.67% of counterfeits are recognised. The border, which holds the note's margins but often also the table or hand beneath it, thus suffices to rank most photographs. Because genuine and counterfeit notes came from different sources (banks and RAB), capture conditions may contribute to decisions [2].

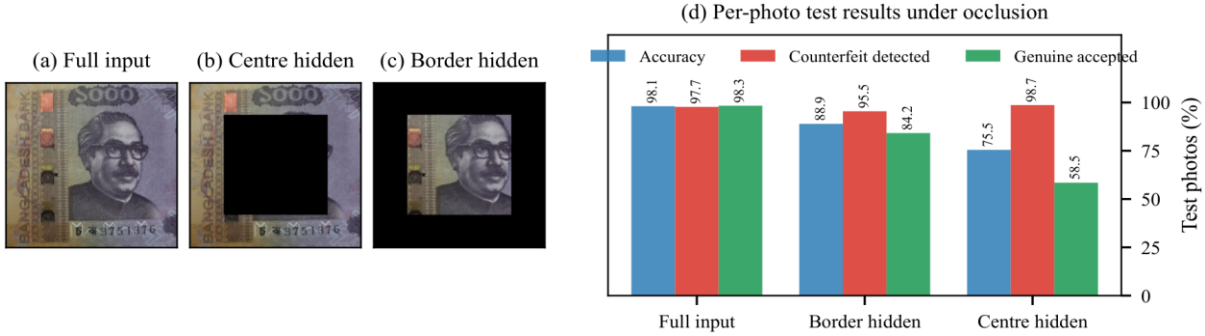


Fig. 8. Occlusion diagnostic. (a)–(c) Inputs with the centre or the 20% border hidden. (d) Per-photograph test results under each condition.

4.6. Benchmark Audit: Duplicates and Label Conflicts

The perfect agreement between a note's photographs, the strong frozen baselines, and the documented augmentation led us to search for duplicates. For every pair of notes, we computed the normalised cross-correlation (NCC) of 48×48 grey-scale thumbnails of corresponding views, averaged over the six views and maximised over flips. The best-match distribution is bimodal (Fig. 9(a)): for validation and test notes, no match lies between 0.963 and 0.997. We therefore grouped notes with NCC > 0.98 (Table 3).

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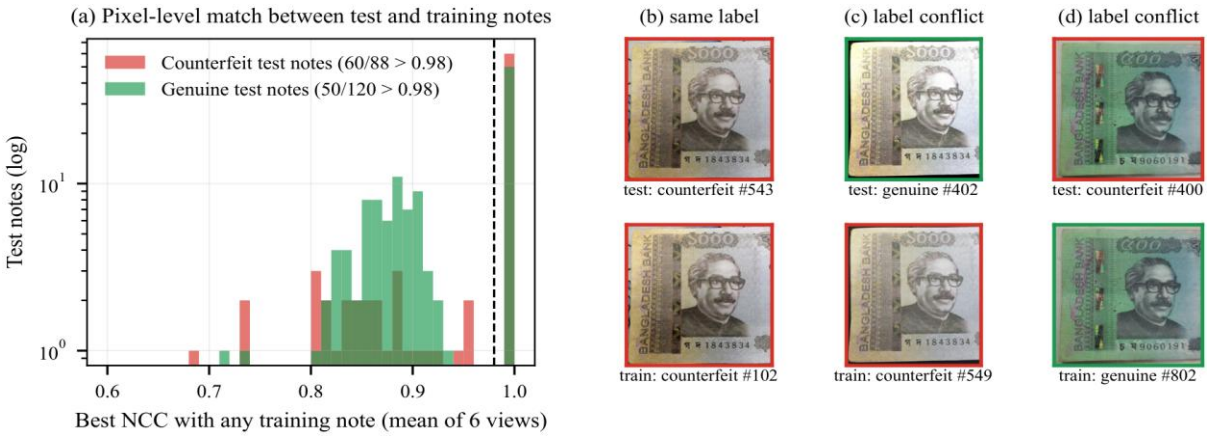


Fig. 9. Duplicate audit. (a) Best NCC match of each test note to any training note; dashed line: threshold 0.98. (b)–(d) A test note (top) and its matching training note (bottom); green frame: genuine, red: counterfeit. (b) Augmented copy, same label. (c), (d) Identical photographs with opposite labels.

Table 3. Duplicate audit of JaalTaka (pixel NCC > 0.98 over six views)

Quantity	Value
Notes / distinct photograph sets (groups)	1,390 / 846
Groups of size 1 / 2 / 3	449 / 250 / 147
Notes sharing their photographs with another note	941 (67.7%)
Groups with only counterfeit / only genuine labels	289 / 545
Groups containing *both* labels	12 (31 notes)
Test notes with a copy in training or validation	116 of 208 (55.8%)
counterfeit / genuine	60 of 88 / 56 of 120
Test notes in label-conflict groups	6
Test errors explained by label conflicts	4 of 4

The augmented copies were thus stored as independent notes, and a note-level split places copies of 55.8% of test notes in training or validation. Twelve groups contain genuine- and counterfeit-labelled copies of the *same* photographs (e.g., genuine notes 400–406 duplicate counterfeit notes 106–112 and 547–553, and counterfeit notes 400 and 403–406 duplicate genuine notes 802 and 1–4), so at least one label per group is wrong; genuine-labelled notes 402 and 403 carry serial number 1843834, which also appears on counterfeit-labelled notes. *All four* test errors are notes from these groups, each predicted with the label of its training copy. On the 92 *clean* test notes, whose group has no member in training or validation, and on the 202 notes outside label-conflict groups, all three methods are error-free (Table 4). The 98.08% accuracy is thus limited by label noise, not by the model, and the task is saturated even for 1-NN on generic features.

Table 4. Note-level accuracy (%) on audited test subsets

Test subset	All	Without label conflicts	Clean (no copy)
Notes (genuine / counterfeit)	208 (120/88)	202 (117/85)	92 (64/28)
1-NN, frozen features	97.12	100.0	100.0
Linear probe, frozen	98.08	100.0	100.0
Ours (fine-tuned)	98.08	100.0	100.0
95% Wilson CI	95.16–99.25	98.13–100	95.99–100

5. Discussion

Within JaalTaka, the system is error-free once label conflicts are removed (lower 95% bound 96.0% on the 92 clean notes), its errors are fully explained, and it runs in real time on a CPU. This does not show that it will detect *new* counterfeits: the 588 counterfeits collapse into 289 photograph sets, those we inspected share a few serial numbers, and they were collected separately from the genuine notes. A model, even 1-NN, can therefore succeed by recognising known batches or capture sessions, as the occlusion test suggests. The saturated benchmark also cannot rank methods, since differences between published CNNs are smaller than its label noise.

We recommend (i) removing or relabelling the 12 conflict groups; (ii) splitting by duplicate group, for which we release our list; (iii) holding out counterfeit families (e.g., by serial number or seizure batch) and capture devices, to measure generalisation to unseen

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forgeries and phones; and (iv) reporting confidence intervals and frozen-feature baselines, which immediately reveal saturation.

5.1. Limitations

The system covers only Tk 500 and Tk 1000. Visible-light photographs cannot capture ultraviolet, magnetic, or tactile features, so the output is decision support and suspicious notes should be confirmed at a bank. Scans and distant photographs are mostly rejected, the OOD evaluation is preliminary, and probabilities are not calibrated on independent data [25]. Retraining on a group-aware split with a counterfeit-family hold-out is the most important next step.

6. Conclusion

This study presented a smartphone-based authenticator for Bangladeshi Tk 500 and Tk 1000 notes that combines a fine-tuned EfficientNet-B0, fusion of up to six photographs, a three-way decision with adjustable strictness, an out-of-distribution gate, and Grad-CAM explanations. The system runs in real time on a CPU and reaches 98.08% note-level accuracy (ROC-AUC 0.994) on the JaalTaka benchmark. Beyond accuracy, the main contribution is an audit of the benchmark itself: its 1,390 notes form only 846 distinct photograph sets, and label conflicts among duplicated notes explain every test error. On test notes without a copy in training, the proposed system and two simple frozen-feature baselines are all error-free, which shows that the benchmark is saturated and can no longer rank methods. In practice, the system is best used as decision support, with suspicious notes confirmed at a bank. The study is limited to two denominations and visible-light photographs, and its out-of-distribution test is preliminary. Future work should retrain and evaluate the system on duplicate-group splits that hold out counterfeit families, capture sessions, and devices.

Declarations

Funding: This research received no external funding.

Competing interests: The authors declare no competing interests.

Ethics approval and consent to participate: Not applicable. This study involved no human participants, personal data, or animals; it used only banknote photographs from a public CC BY 4.0 dataset [26].

Consent for publication: Not applicable.

Data availability: JaalTaka is available from Mendeley Data [26]. The duplicate-group list is available from the corresponding author and will be released publicly upon publication.

Materials availability: Not applicable.

Code availability: The model and code are available from the corresponding author and will be released publicly upon publication.

Use of generative AI: The authors used an AI assistant to help write analysis scripts and edit the text, and checked all code, results, and references.

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